VERY IMPORTANT QUESTIONS UNIT 1-5

Basic Electrical and Electronics Engineering

Comprehensive Study Notes

Units 1 - 5

UNIT - 1

1ST PART

- Problems on current and voltage division.
- Problems on resistor simplification.
- Define Ohm's law and explain its limitations.
- Define KVL and KCL and explain with examples.

2ND PART

- Define and derive the expression for RMS value, Average value, Form factor, and Peak factor.
- What is Power factor? Explain the importance of Power factor.
- Draw the Power triangle and explain the formulas.
- Explain: Time period, Peak value, Frequency, Peak-to-peak value, and Amplitude.

UNIT - 2

1ST PART

- Explain the principle and operation of a Single-phase Transformer.
- Explain the principle and operation of a DC Generator.
- Explain the principle and operation of a 3-phase Induction Motor.
- Explain any two types of Single-phase Induction Motors.
- Explain the principle and operation of a DC Motor (with diagram).

2ND PART

- Explain the operation of Wheatstone Bridge and derive the expression for an unknown resistor.
- Explain the operation of Megger with a neat diagram.
- Explain the principle and operation of PMMC instrument with a neat diagram.
- Explain the principle and operation of MI (Moving Iron) repulsion type instrument with a neat diagram.
- Explain the operational differences between MC (Moving Coil) and MI (Moving Iron) instruments.

UNIT - 3

1ST PART

- Explain the layout diagram of a Thermal Power Plant and explain each and every part.
- Draw a Hydroelectric Power Plant and explain each and every part.
- Draw the layout diagram of a Wind Plant and explain each and every part.
- Draw the layout diagram of a Solar Power Plant and explain each and every part.

2ND PART

- Problems on Energy Meter.
- What is electric shock? Explain the safety precautions to avoid electric shock.
- Explain the operation of a Fuse and MCB (Miniature Circuit Breaker).
- Explain the different types of earthing.

UNIT - 4

1ST PART

- Explain the operation of P-N junction diode and draw the V-I characteristics.
- Explain the operation of Zener diode and draw V-I characteristics.
- Explain how a Zener diode acts as a Zener voltage regulator.

2ND PART

- Explain the operation of N-P-N and P-N-P transistors.
- Explain Common Base (CB) mode of transistor; draw input and output characteristics.
- Explain Common Emitter (CE) mode of transistor and draw input/output characteristics.
- Analyze the performance of CB, CE, and CC modes of transistors.

UNIT - 5

Analog Electronics

- Explain Half-wave rectifier and Full-wave rectifier. Draw the waveform and derive the expression for output voltage.
- Explain the operation of a Bridge rectifier and draw the waveform; derive expressions for output voltage and output current.
- Explain different types of filters and draw the corresponding waveforms.

Digital Electronics

- Explain different types of logic gates with truth tables and logic diagrams.
- Explain Half adder and Full adder with logic diagrams and truth tables.
- Problems on Hamming Code.
- Explain Excess-3 code and Gray code (from 0 to 10).
- Octal and Hexadecimal code conversion: $[256]_{16} \rightarrow$ Decimal (Sums like DLD).